

4.3 Video Codecs and Containers

The need to transfer digital data across a network brought about the need for codecs. Put simply, ‘codec’ stands for ‘Compression-Decompression’, or ‘Coding-Decoding’. A video codec is a software module that enables the compression and decompression of digital video. As video technology progressed, storing and using video in digital format became feasible, and a variety of such technologies began to emerge. Video compression required customised compression techniques. Care had to be taken to ensure there was no loss of quality during compression. This led to the development of algorithms that compressed the software. Though different algorithms and methods have been devised to compress and decompress data, a few factors are considered cardinal rules. The compression algorithm should be robust enough to ensure minimal data loss and errors during compression. Ease of editing the file after compression and random access are a couple of other factors that need to be considered while designing a video codec.

With the number of codecs available growing every day, it has become necessary to bundle these into ‘packs’, or containers. Most of the codec packs available today support a multitude of codecs, and also have players and other video utility tools bundled along.

In addition to codecs we also have something known as container formats. A container is a file format that can store various types of data in it. This data is compressed in a uniform manner using a standard codec. Advanced container formats can therefore support audio, video, subtitles, and meta-data. They are also capable of storing the synchronisation information needed to play back the various data streams together, as in a typical video playback. Thus one of the essential differences between a codec pack and a container is that the latter acts as an envelope for different media streams. A few of the better known container formats are Apple’s QuickTime and Microsoft’s Advanced Streaming Format (ASF).